

Unitary in the Equations, Measurement in the Reductions

A paired-message rho calculus with a value binder,
path-sum equations, and traced execution

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Working note, 1 October 2026

Abstract

This note records a design session exploring a single intuition: in a graph-structured lambda theory the *unitary* behaviour lives in the (possibly augmented) equations, while *measurement* lives in the reductions. The rho calculus with graded where-clauses is the vehicle, because every weighted GSLT translates into it. Working the intuition out forces a small number of changes to rho: messages and names become *pairs* of a process and a phase label; dropping splits into a binary, phase-preserving *equation* and a unary, semantically-substituted *execution* whose erasure is *traced*; superposition enters through a binder Σ over $\mathbb{Z}/2$ -valued variables; and couplings between values are inert atoms governed by path-sum equations, which are the algebraic face of the graph-like ZX-calculus. Reductions then come in exactly two measuring flavours (resolving a value, or running received code) and one coherent flavour (binding receipts). Persistent receipts are dropped, recursion is recovered by reflection, and loop guards are classical by theorem. A worked example computes $HZH|0\rangle = |1\rangle$ by equations alone, and contrasts it with a which-rail detector that destroys the interference by reductions alone. The note closes with the decisions taken and the questions left open.

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1 The intuition and where it comes from

The intuition under study is:

The unitary behaviour of a quantum-interpreted GSLT lives in its (possibly augmented) equations; measurement lives in its reductions.

It has roots in earlier work in this line. The first version of the weighted-GSLT note assigned the Hamiltonian to equations and the jump operators to rewrites, and its second version distilled the slogan that a weighted GSLT is quantum exactly to the extent that its presentation withholds equations from the quotient. What is new here is taking the intuition as the *organising principle* for ρ with graded where-clauses. That reverses the direction of the graded-where programme [13], in which reductions carried coherent amplitude: here reductions are where coherence ends.

Three observations make the reversal natural in ρ .

- **COMM quotes.** Firing substitutes $@Q$ into a continuation; the payload becomes a name, and names are compared by equality. A communication therefore writes a record, which is precisely Barandes’s criterion for a division event [3].
- **Cost.** Equations are free and rewrites are costed. Cost lands where information is erased, which is Landauer’s principle read operationally.
- **Hermiticity.** An equation is symmetric; a directed rewrite is not. A weighted symmetric relation can be Hermitian; a directed one cannot.

Reading order. Section 2 retraces the design path, because several of the choices only make sense against the alternatives that were rejected. Section 3 states the calculus. Section 4 records its basic properties. Section 5 works an example. Sections 6 and 7 place the work and list what remains.

2 The design path

2.1 Embedding ZX: the graph-like fragment

The starting point was to embed the ZX-calculus [6] into ρ , and the cleanest target is the *graph-like* fragment [8]: only green spiders, joined by Hadamard edges, into which every diagram rewrites and to which PyZX normalises. Reading a quantum name as a Z -spider, scope as the boundary, and AC of parallel composition plus scope extrusion as “only connectivity matters” gives spider fusion and identity removal for free.

Two lessons came from this first pass and survive into the final design.

1. **Once the congruence contains a complete ZX theory, syntactic matching on quantum structure is not well defined.** The $\equiv$ -class of a quantum subterm is its linear map, so the only $\equiv$ -invariant observations are semantic. Deciding $\equiv$ is simulation.
2. **Unbinding is post-selection.** Making a spider internal plugs $\langle 0| + \langle 1| = \sqrt{2}\langle +|$ into it. Genuine discard needs the ground of $ZX_{\perp}$ [5], and that is where reductions belong.

2.2 Quoted pairs, and closure under reflection

The proposal that channels be quoted pairs $@(P, \phi)$ moves the phase into the name. Making only channels into pairs breaks the invariant that keeps ρ closed under reflection: the payload sort and the quoted sort must coincide. Making *messages* pairs as well, $x!(Q, \phi)$, restores it. A communication substitutes $@(Q, \phi)$, which is again a name.

2.3 Binary drop

The remaining seam was the drop in process position, which would yield a pair where a process is expected. The resolution is a *binary* drop

$$*(@(P, \phi), u) := u!(P, \phi),$$

which never produces a process. It is invertible (the message determines both arguments), it preserves the phase, and it is an equation. On its own it gives an exactly closed calculus, but nothing executes quoted code any more, so recursion is lost; with linear receipts every term strongly normalises.

2.4 Semantic substitution and traced erasure

Recursion is recovered as in the original ρ paper [12]: drop in process position is resolved by *semantic substitution*, and on substitution the process begins running and its phase is erased.

Erasure read literally, $(P, \phi) \mapsto P$, is many-to-one; amplitudes of branches that differ only in label add, which is not an isometry and reintroduces the norm leak (and with it PostBQP). Erasure is therefore read as a *trace*: the information goes to an environment nobody reads, and the step is a completely positive instrument. Substitution is *eager*: the trace happens at the communication, even if the drop is guarded and never reached.

What the trace records was refined twice. Recording only the label lets superposed control through (two routed payloads with equal labels but different code). Recording the *whole message* closes the gap, and is the choice most consistent with context-labelled bisimulation [11], where what an observer learns from a transition is its label and labels carry whole contexts. Messages are compared up to structural congruence.

2.5 The coupling: bicharacter, and by equation

At a communication, a sending label a and a receiving label b could combine as a difference $e^{i(a-b)}$ or as a bicharacter $\chi(a, b)$. The difference is a $U(1)$ connection with gauge freedom per channel, and it has a sharp limitation: every maximal matching of a complete join uses each occurrence once, so all matchings carry the same total phase and the site matrix has rank one. The bicharacter is what ZX is built on. Hadamard has entries $(-1)^{ab}/\sqrt{2}$, a Hadamard edge in graph-like ZX is a CZ with amplitude $(-1)^{ab}$, and a Z -spider of phase $k\pi/4$ contributes ω^{ka} with $\omega = e^{i\pi/4}$, which is a bicharacter $\mathbb{Z}/8 \times \mathbb{Z}/2 \rightarrow U(1)$. In ZX the coupling is static, so it is evaluated by *equation*, not at the communication.

2.6 A correction: labels are data, couplings are atoms

A static coupling needs values to sum over, hence a binder. The first attempt put value-dependent terms into the phase slot of a pair. That fails for two reasons.

- A label that depends on a summed variable is *data in the configuration*, so the branches it separates are distinct basis configurations. It is a which-path record, and it destroys the very interference the coupling was meant to produce.

- The trace reads a message's label, and the path-sum equations rewrite labels, so the reduction would not be invariant under $\equiv$.

The resolution separates the two roles. The phase slot of a pair holds a *constant* $k \in \mathbb{Z}/8$, which is data: part of the name, and traced on execution. Couplings between values live in a separate inert atom $\langle\langle\phi\rangle\rangle$ that is never received, run or traced, so the equations may rearrange it freely.

3 The calculus

3.1 Syntax

Definition 3.1 (Syntax).

$e ::= v \mid 0 \mid 1 \mid e \oplus e$	values, in $\mathbb{Z}/2$
$\phi ::= k \cdot m \mid \phi + \phi$	$k \in \mathbb{Z}/8$, m a monomial in values
$s \in \mathbb{Z}[\frac{1}{\sqrt{2}}, \omega]$	scalars, $\omega = e^{i\pi/4}$
$M ::= (P, k)$	messages, $k \in \mathbb{Z}/8$ constant
$x ::= @M \mid y \mid x[e]$	names
$P ::= 0 \mid x!M \mid \text{for}(y \leftarrow x)P \mid \text{for}(y \leftarrow x[u])P$ $\mid *y \mid *(x, u) \mid P \mid P \mid \Sigma v. P \mid \langle\langle\phi\rangle\rangle \mid s \cdot P$	processes

The receipt $\text{for}(y \leftarrow x[u])P$ binds both y and the index variable u . $\Sigma v. P$ binds the value variable v in P , including occurrences in indices and couplings. For Clifford, monomials of degree at most two suffice; $(-1)^{ab}$ is written $\langle\langle 4 \cdot ab \rangle\rangle$.

Definition 3.2 (Name identity). Indexed names $x[0]$ and $x[1]$ are distinct. A name $@(P, k)$ is identified by its whole pair; the label k is part of the name, as data. A name $x[e]$ with e depending on Σ -bound variables denotes different channels in different branches.

Remark 3.3 (What is not primitive). There are no persistent receipts, no replication and no summation operator on processes. There is no restriction operator either; fresh names arise by quotation. The binder Σ is deliberately separate from any freshness binder: Σ gives a spider a value, a fresh name gives a channel an identity, and conflating them would make every fresh channel a superposition.

3.2 Equations: the unitary side

Definition 3.4 (Structural congruence). $\equiv$ is the least congruence, closed under α -conversion of both name and value binders, satisfying:

(E1) Monoid and scalars

$$P \mid 0 \equiv P, P \mid Q \equiv Q \mid P, (P \mid Q) \mid R \equiv P \mid (Q \mid R); (s \cdot P) \mid Q \equiv s \cdot (P \mid Q), \\ s \cdot (t \cdot P) \equiv (st) \cdot P, 1 \cdot P \equiv P.$$

(E2) Binary drop

$$*(@M, u) \equiv u!M.$$

(E3) Scope

$$(\Sigma v. P) \mid Q \equiv \Sigma v. (P \mid Q) \text{ when } v \notin \text{fv}(Q); \Sigma v. \Sigma w. P \equiv \Sigma w. \Sigma v. P; \Sigma v. (s \cdot P) \equiv s \cdot \Sigma v. P.$$

(E4) Couplings

$$\langle\langle\phi\rangle\rangle \mid \langle\langle\psi\rangle\rangle \equiv \langle\langle\phi + \psi\rangle\rangle, \langle\langle 0 \rangle\rangle \equiv 0, \langle\langle k \cdot 1 \rangle\rangle \equiv \omega^k \cdot 0, \text{ with arithmetic in } \mathbb{Z}/8 \text{ and } v^2 = v.$$

(E5) Elim

$\Sigma v. P \equiv 2 \cdot P$ when $v \notin \text{fv}(P)$.

(E6) HH

$\Sigma w'. \Sigma w. (\langle\langle 4 \cdot w \cdot (w' \oplus e) \rangle\rangle \mid P) \equiv 2 \cdot P\{e/w'\}$ when $w \notin \text{fv}(P, e)$ and $w' \notin \text{fv}(e)$.

(E7) ω -rule

The rule completing the Clifford fragment of the path-sum calculus [1]; its statement in this syntax is open (Question 7.4).

E6 is the identity $\sum_{w \in \mathbb{Z}/2} (-1)^{wf} = 2\delta_{f=0}$: a linear constraint on a summed variable becomes a substitution. In ZX it is the Hopf law and Hadamard cancellation; in this calculus it is name fusion. E2 is the only equation that touches communication syntax, and it is a bijection between term shapes. Every equation is an equality, hence invertible, so the unitary content sits where the intuition places it.

3.3 Denotation

Definition 3.5 (Path-sum denotation). Every closed term is $\equiv$ to one of the form $\Sigma \vec{v}. (s \cdot \langle\langle \phi \rangle\rangle \mid C)$ with C coupling-free and scalar-free. It denotes

$$\llbracket \Sigma \vec{v}. (s \cdot \langle\langle \phi \rangle\rangle \mid C) \rrbracket = s \sum_{\vec{v} \in \mathbb{Z}/2^n} \omega^{\phi(\vec{v})} |C(\vec{v})\rangle,$$

where the configurations $C(\vec{v})$, taken modulo E1–E3, form an orthonormal basis.

This is the path-sum representation of Dawson et al. [7] and Amy [1]: the algebraic twin of ZX. Note that in this architecture a term class *is* a vector; there is no time evolution generated by the equations, and gates are applied by composition.

3.4 Reductions: the measurement side

Reductions are closed under parallel composition, scalars, Σ , and $\equiv$. Write $\text{idx}(c[e]) = e$ and $\text{idx}(x) = 0$ for an unindexed name.

Definition 3.6 (Reduction rules). **(R1) Coherent COMM**

The rule is

$$c[e]!(P, k) \mid \text{for}(y \leftarrow c[u])Q \rightarrow \langle\langle k \cdot e \rangle\rangle \mid Q\{\text{@}(P, k)/y, e/u\}.$$

The receipt binds the index symbolically, so nothing is resolved. The payload's label is *banked* as amplitude against the index: this is the bicharacter coupling of a phase with a value. The unindexed case, $x!(P, k) \mid \text{for}(y \leftarrow x)Q \rightarrow Q\{\text{@}(P, k)/y\}$, banks nothing.

(R2) Resolving COMM

A receipt on $c[b]$ with b a constant, facing $c[e]!M$ with e dependent on Σ -bound variables, fires only on the assignments where $e = b$. The step is a projective instrument with two outcomes: *fired*, which restricts the sum to $e = b$ and then proceeds as R1; and *not fired*, which restricts to $e \neq b$. Outcome probabilities are Born weights of the restricted sums. In general: *a communication whose enabledness varies across assignments measures the variables it depends on.*

(R3) Traced execution

If $*y$ occurs in process position anywhere in Q (guarded occurrences included), the communication additionally records the received message $M = (P, k)$, up to $\equiv$, to an environment, and

$$(*y)\{\text{@}(P, k)/y\} = P.$$

On binary drops the substitution is syntactic: $*(y, u)\{\textcircled{M}/y\} = *(\textcircled{M}, u) \equiv u!M$.

Remark 3.7 (R3 as an instrument). With Kraus operators indexed by $\equiv$ -classes of messages, $K_{[M]}$ projects onto the branches in which M was consumed and then substitutes; $\sum K^\dagger K = I$, so the step is trace preserving. Its Stinespring form writes M to a channel nobody holds. Three perspectives coincide with those requested of the heralds in [13]: globally the state is pure; for anyone without the record’s capability it is dephased; for anyone who reads the record it is a measurement with an outcome.

Remark 3.8 (When the trace is inert). The record is inert, and the step coherent, exactly when the same message (up to $\equiv$) is consumed in every branch. In particular running code with a constant label and no Σ -bound values in it costs no coherence.

4 Properties

Proposition 4.1 (Closure under reflection). *Quotation takes messages to names and the binary drop takes names back to messages: $*(\textcircled{M}, u) \equiv u!M$, and every communication substitutes a name of the form $\textcircled{M}$. The payload sort and the quoted sort coincide.*

Proposition 4.2 (Replication is derivable). *Let $R := \text{for}(y \leftarrow x)(*(y, x) \mid *y)$ and $!P := x!(R \mid P, 0) \mid R$. Then $!P \rightarrow x!(R \mid P, 0) \mid R \mid P$. The trace at each unfolding records the same message in every branch and is therefore inert.*

Proposition 4.3 (Classical control). *Within the scope of a Σ binder, which communications fire and which code runs is the same under every assignment of the bound values, except at R2 and R3 events.*

Proof sketch. Σ -bound values occur only in indices, couplings and inside quoted payloads. A message’s presence cannot depend on a value except through its index, so unindexed communications are enabled uniformly, and binding receipts $c[u]$ fire uniformly. A communication whose enabledness varies is R2, which resolves; this includes matching a quoted name whose identity depends on a value. The code that runs can differ across branches only if the messages consumed differ, and R3 records them. $\square$

Remark 4.4 (Consequences of Proposition 4.3). The coherent part of any run is a fixed circuit whose shape is classical. This is Selinger’s *quantum data, classical control* [15]. Loop guards are classical, so loops are monotone and the Löwner-order obstruction to recursion under quantum alternation [2] does not arise. Mach–Zehnder style routing survives, because it moves data along rails without forking code, so the positioning against quantum alternation in [13] is unaffected.

Remark 4.5 (The discipline the trace imposes). Running code that mentions a Σ -bound value measures that value, because a context could tell the variants apart. Quantum data must therefore enter code through binding receipts, never through the code itself. Gates written as in Section 5 obey this: their code mentions only their rails, and they acquire the index at the communication.

Remark 4.6 (Measurement criterion). A communication is coherent when its subject does not depend on a bound value (or is matched by a binding receipt) *and* its continuation does not run what it receives (or runs the same message in every branch). Otherwise it measures. This is a weak reading of “measurement in the reductions” with a syntactic criterion; the strong reading is the special case in which every continuation runs what it receives.

5 A worked example

A qubit is a token $T := 0$ on a rail $q[v]$. A Hadamard gate is a process:

$$H(q, q') := \text{for}(y \leftarrow q[u]) \Sigma w. (2^{-1/2} \cdot \langle\langle 4 \cdot u \cdot w \rangle\rangle \mid *(y, q'[w])).$$

It binds the input index u , sums a fresh output value w with coupling $(-1)^{uw}$, and forwards the pair unchanged by binary drop. Nothing is run, so the gate is coherent.

Example 5.1 ($HZH|0\rangle = |1\rangle$ by equations alone). Start from

$$q_0[0]!(T, 4) \mid H(q_0, q_1) \mid H(q_1, q_2).$$

The token carries label 4, so every hop banks $(-1)^{\text{index}}$; it acts as a Z at each hop.

1. *Fire* $H(q_0, q_1)$. By R1, $u := 0$ and $\langle\langle 4 \cdot 0 \rangle\rangle \equiv 0$:

$$2^{-1/2} \cdot \Sigma w. q_1[w]!(T, 4) \mid H(q_1, q_2),$$

which denotes $|+\rangle$.

2. *Fire* $H(q_1, q_2)$. After extruding Σw over the gate (E3), R1 binds $u := w$ symbolically and banks $\langle\langle 4 \cdot w \rangle\rangle$, which is the Z :

$$2^{-1/2} \cdot \Sigma w. (\langle\langle 4 \cdot w \rangle\rangle \mid 2^{-1/2} \cdot \Sigma w'. (\langle\langle 4 \cdot w \cdot w' \rangle\rangle \mid q_2[w']!(T, 4))).$$

3. *Equations only*. By E4 the couplings fuse to $\langle\langle 4 \cdot w \cdot (w' \oplus 1) \rangle\rangle$; by E1 and E3 the scalars float out and the binders reorder to $\Sigma w'. \Sigma w$; E6 with $e = 1$ gives

$$2^{-1/2} \cdot 2^{-1/2} \cdot 2 \cdot q_2[1]!(T, 4) = q_2[1]!(T, 4).$$

The $w' = 0$ branch is cancelled by interference inside the equations, with no reduction involved. A receipt on $q_2[1]$ now fires with probability 1. With label 0 the same steps give $q_2[0]!(T, 0)$, which is $HH = I$.

Example 5.2 (A which-rail detector destroys the interference). Replace the second gate's input rail by detectors:

$$\text{for}(y \leftarrow q_1[0])*(y, d[0]) \mid \text{for}(y \leftarrow q_1[1])*(y, d[1]) \mid H(d, q_2).$$

Each detector is enabled on only one value of w , so by R2 each resolves w . The state becomes an equal mixture of $d[0]!(T, 4)$ and $d[1]!(T, 4)$, and the gate then yields each output with probability $1/2$. The equations are unchanged; only the reductions differ, and they alone remove the interference.

6 Context

ZX and path sums. The equations are path-sum rewrite rules [7, 1], whose completeness for Clifford is known, and which are the algebraic counterpart of graph-like ZX [8] with its completeness results [10, 16]. R3 plays the role of the ground in $\text{ZX}_\perp$ [5].

Measurement-based computation. Graph states with local phases plus single-qubit measurements and classical feed-forward are universal [14], and gflow [4] is the condition under which internal spiders are realised by real measurements and corrections rather than post-selection. The calculus has this shape: static couplings, measuring reductions, classical control. This suggests that a strong reading of the intuition costs no expressiveness, but that has not been checked here.

Quantum control versus quantum data. The classical-control theorem places the calculus with Selinger [15] and with CQP [9] as regards control, while the coherent routing of R1 keeps the Mach–Zehnder reading of quantum alternation [2, 17] available for data.

Barandes. Division events [3] are R2 and R3 steps: moments at which the branch distinction is written into a name or a record. Between them the process is indivisible in his sense.

Context-labelled bisimulation. R3 records exactly what a context-labelled transition exposes, so decoherence classes and observation classes coincide by construction (up to the $\equiv$ -versus- $\sim$ gap noted in Question 7.5).

The catamorphism notes. The weighting notes in the repository dilate the rewrite operator and post-select per step. This note moves the unitary out of the rewrites altogether; if the weighting functor is required to respect the equations, augmentation makes it a projective (twisted) functor.

7 Decisions and open questions

7.1 Decisions taken

Question	Decision
Payload sort	Messages are pairs (P, k) ; names are quoted messages.
Drop	Binary drop is an equation; unary drop is semantic substitution.
Erasure on execution	Traced, not coherent.
When the trace happens	Eagerly, at the communication.
What the trace records	The whole message, up to $\equiv$.
How labels couple	By bicharacter, evaluated by equation.
Label versus coupling	Labels are constant data; couplings are inert atoms.
Superposition	A dedicated binder Σ over $\mathbb{Z}/2$.
Persistent receipts	Dropped; replication is derivable.
Loop guards	Classical (now a theorem).

7.2 Open questions

Open question 7.1 (Per-hop banking). As written, R1 banks a payload’s label at every indexed channel it crosses, so a labelled payload acts as a phase gate at every hop. This is consistent but may be more than intended. The alternative is to bank once, for instance when the label is first placed.

Open question 7.2 (Born normalisation). How normalisation interacts with maximal progress, and whether R2’s “not fired” outcome should be an outcome at all or be absorbed by maximal progress.

Open question 7.3 (Race normalisation). Re-deriving contention-set completion [13] in this calculus, now that some communications are instruments and the coherent ones only move pairs.

Open question 7.4 (E7 and the zero scalar). Stating the ω -rule in this syntax, and deciding how to treat $0 \cdot P$, which collapses all terms of a given shape.

Open question 7.5 ($\equiv$ versus $\sim$). The trace compares messages up to $\equiv$, which can decohere branches that no context distinguishes. Coarsening to bisimilarity is available later, at the price of a fixed-point definition.

Open question 7.6 (Beyond Clifford). Completeness of the equations once T-type couplings are present, and the cost of comparing payloads that carry their own internal sums in R3.

Acknowledgement. The design was worked out in dialogue with Claude (Anthropic), which also drafted this note.

References

- [1] M. Amy. Towards large-scale functional verification of universal quantum circuits. *QPL 2018*, EPTCS 287, 2019.
- [2] C. Bădescu and P. Panangaden. Quantum alternation: prospects and problems. *QPL 2015*, EPTCS 195, 2015.
- [3] J. A. Barandes. The stochastic-quantum correspondence. arXiv:2302.10778, 2023.
- [4] D. E. Browne, E. Kashefi, M. Mhalla and S. Perdrix. Generalized flow and determinism in measurement-based quantum computation. *New J. Phys.* 9:250, 2007.
- [5] T. Carlette, E. Jeandel, S. Perdrix and R. Vilmart. Completeness of graphical languages for mixed states quantum mechanics. *ICALP 2019*.
- [6] B. Coecke and R. Duncan. Interacting quantum observables: categorical algebra and diagrammatics. *New J. Phys.* 13:043016, 2011.
- [7] C. M. Dawson, H. L. Haselgrove, A. P. Hines, D. Mortimer, M. A. Nielsen and T. J. Osborne. Quantum computing and polynomial equations over the finite field $\mathbb{Z}_2$. *Quantum Inf. Comput.* 5(2):102–112, 2005.
- [8] R. Duncan, A. Kissinger, S. Perdrix and J. van de Wetering. Graph-theoretic simplification of quantum circuits with the ZX-calculus. *Quantum* 4:279, 2020.
- [9] S. J. Gay and R. Nagarajan. Communicating quantum processes. *POPL 2005*.
- [10] E. Jeandel, S. Perdrix and R. Vilmart. A complete axiomatisation of the ZX-calculus for Clifford+T quantum mechanics. *LICS 2018*.
- [11] J. J. Leifer and R. Milner. Deriving bisimulation congruences for reactive systems. *CONCUR 2000*.
- [12] L. G. Meredith and M. Radestock. A reflective higher-order calculus. *ENTCS* 141(5):49–67, 2005.
- [13] L. G. Meredith. Races decided by the Born rule (graded-where, quantum case, version 6). F1R3FLY-io/publications, **fuzzyware**/, 2026.
- [14] R. Raussendorf and H. J. Briegel. A one-way quantum computer. *Phys. Rev. Lett.* 86:5188, 2001.
- [15] P. Selinger. Towards a quantum programming language. *Math. Struct. Comput. Sci.* 14(4):527–586, 2004.

- [16] R. Vilmart. A near-minimal axiomatisation of ZX-calculus for pure qubit quantum mechanics. *LICS 2019*.
- [17] M. Ying, N. Yu and Y. Feng. Alternation in quantum programming: from superposition of data to superposition of programs. arXiv:1402.5172, 2014.